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Integration of building information modelling and digital twin in mechanized tunnelling construction

Eder Martinez and Felix Stauch
Division Civil Engineering, Implenia Switzerland AG, Glattpark (Opfikon), Switzerland
Borja García de Soto
Division of Engineering, New York University – Abu Dhabi Campus, Abu Dhabi, United Arab Emirates, and
Rafael Sacks
Faculty of Civil and Environmental Engineering, Technion Israel Institute of Technology, Haifa, Israel

Abstract

Purpose – This study explores the integration of monitored data from Tunnel Boring Machine (TBM) operations into a digital twin framework to generate real-time status information for mechanized tunnelling. The study investigates the potential benefits that can be derived in terms of lean operations and reduced waste of rework.
Design/methodology/approach – Using a design science research approach within a case study of a tunnelling project, the researchers developed, tested and refined the Digital Twin Construction (DTC) system and observed use cases of its application. The empirical data collected was analyzed to draw conclusions concerning the feasibility of the integration of TBM excavation data into a digital twin and the impacts of its use in improving flow and avoiding rework by reducing uncertainty.
Findings – The study identifies key technical challenges, such as imperfect data interoperability across different systems involved in tunnelling operations and the need for standardized approaches to modelling. It also highlights the value of the digital twin in supporting lean design and construction workflows by providing the project team with information drawn from a dynamic representation of the tunnel's physical state in short cycle times.
Originality/value – This study presents an innovative computational design algorithm that automates data integration, enabling digital twin generation as a functional product. Furthermore, it includes use cases illustrating how the resulting digital twin supports the improvement of specific operations, such as bracketry design and construction. By addressing both technical and operational dimensions, this case study serves as a valuable resource for researchers and industry practitioners aiming to advance digital twin technology and lean construction in tunnelling and broader construction applications.

Keywords Building information modelling, Digital twin, Lean construction, Parametric design, Mechanized tunnelling, Tunnel boring machine, Digital twin construction

Paper type Research paper

1. Introduction

Mechanized tunnel construction using Tunnel Boring Machines (TBMs) generates vast amounts of data, making effective information management critical for project success. In this context, the use of Building Information Modelling (BIM) and digital twin technologies (Parn *et al.*, 2024) offers significant potential to facilitate data integration, real-time monitoring and fact-based decision-making throughout the tunnel lifecycle. Beyond supporting data management, these technologies also align closely with lean construction principles. Many of the positive interactions are related to improving operations flow and reducing the waste of

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rework by providing designers and construction teams with up-to-date, accurate and useful information concerning the current status of the product and processes (Sacks *et al*., 2010).

TBMs are equipped with a rotating circular cutting head to excavate through various types of ground, including rock, soil and mixed conditions. As the cutting head excavates and the TBM advances, the excavated material is conveyed away through an integrated system of conveyor belts, ensuring the continuous progression of the excavation (Maidl *et al*., 2012). Simultaneously, the TBM places precast concrete segments via an automated system, creating a continuous lining behind it that shapes the tunnel structure. Advances in technology have enabled modern TBMs to integrate sophisticated real-time monitoring and sensor systems, generating extensive datasets critical for precise management and optimization of excavation processes (Garcia *et al*., 2021). These data streams include TBM performance metrics, tunnel lining integrity, segment positioning and other operational parameters.

Similarly, advancements in BIM and digital twin technologies have provided the construction industry with powerful means to enhance productivity and streamline processes through improved data integration (Sacks *et al*., 2025), becoming instrumental in enhancing collaboration, efficiency and information management within construction projects (Noor and Yi, 2018; Zhi *et al*., 2018). The evolution of real-time data collection and sensor networks has further facilitated the transition to digital twins (Salzgeber *et al*., 2024), enabling a more dynamic and responsive approach to project management called Digital Twin Construction (DTC) (Sacks *et al*., 2020). In principle, a tunnel digital twin has two parts: (1) a BIM model of the tunnel's design (its *intent* information), and (2) an accurate virtual representation of the tunnel that reflects its physical counterpart in real time (its *status* information). The status information is compiled by integrating various data sources, including sensor data, construction models and operational information (Li *et al*., 2024). This digital twin replica system can be used for monitoring, analysis and decision-making throughout the tunnel's lifecycle, from design and construction to operation and maintenance.

Although earlier studies have examined the use of digital twins for tunnelling (e.g. Wu *et al*., 2022; Yu *et al*., 2021; Zhao *et al*., 2022), there are still several gaps in knowledge that must be addressed. These gaps can be organized into two categories, namely technical and operational. Technical gaps are associated with developing the digital twin system as a functional product. Challenges in this area derive primarily from the complexity of the data generated by TBMs during excavation. Specifically, the sheer volume of real-time data collected from various sensors can overwhelm traditional data processing systems, making it difficult to extract actionable insights (Getuli *et al*., 2021). Moreover, the lack of standardized data formats and interoperability between different systems often hampers effective data sharing among system components, limiting the collaborative potential of digital twins (Stascheit *et al*., 2018).

Operational gaps centre on the effective use of digital twins to inform and support project workflows. This fact is particularly crucial in the context of lean construction, where improving flow, reducing waste and continuous improvement are fundamental goals (Koskela, 1992). Digital twins are not merely virtual models; they represent a dynamic integration of physical assets and digital environments, continuously synchronized through real-time data streams (Grieves, 2014). Their potential lies in their ability to bridge the gap between virtual planning and physical execution, providing a comprehensive, data-driven framework for optimizing construction processes. The seamless integration of diverse project status information into the digital twin is essential for proactively informing lean design and decision-making processes. By enabling real-time monitoring and predictive analytics, digital twins facilitate a more reliable workflow planning, enhance production control and support the systematic elimination of constraints, ultimately improving the quality of flow and reducing waste (Araújo *et al*., 2022). However, a key challenge remains in the completeness and

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accuracy of these models. As noted by Getuli *et al*. (2021), "few constructed facilities have complete as-built information models which gather relevant data besides 3D representations of building components." This gap translates into a limited understanding of how digital twins can be effectively incorporated into real construction processes to fully realize their potential in lean-driven project delivery.

This study investigates the effective integration of TBM-generated data into BIM to form the foundation of a digital twin. By examining a specific tunnelling project where TBM data was systematically captured, processed and utilized to create a dynamic digital replica, it addresses the technical and operational gaps while demonstrating how specific use cases enhance lean tunnel construction. It contributes to knowledge in two key areas. First, it provides insights into the methods for data integration, the challenges faced during the process and formulates the lessons learned from real-world implementation. Second, it analyses use cases where information from previous project phases, such as segment production data, was integrated into the developed digital twin to facilitate related design and production processes, including bracketry design and construction. Ultimately, this case study serves as a valuable reference for industry practitioners and researchers aiming to advance the adoption of digital twin technology in tunnelling and other construction applications.

2. Research design

This study uses the Design Science Research (DSR) methodology to investigate the implementation of BIM and digital twin technologies within a tunnelling project. DSR is a problem-solving paradigm that focuses on the development and evaluation of artefacts intended to address identified issues (vom Brocke *et al*., 2020). This approach is particularly suited for studies that aim to design, implement and assess innovative solutions within complex socio-technical contexts, particularly in fields such as engineering, architecture and information technology (March and Smith, 1995). Accordingly, this research develops and evaluates a digital twin framework tailored to a tunnelling project.

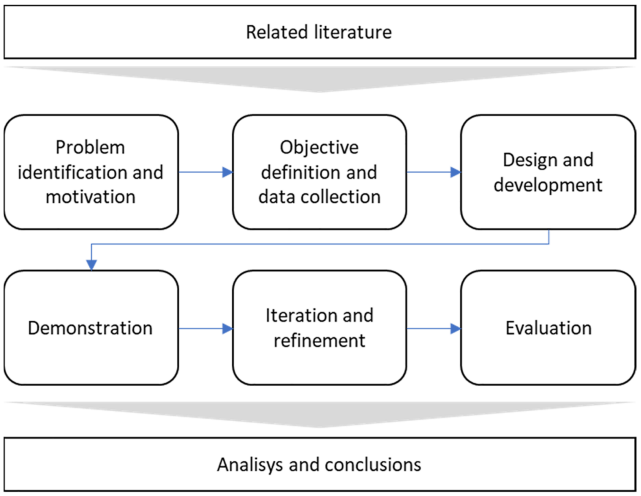
The selection of the case study is based on its representativeness of mechanized tunnelling projects that integrate digital twin methodologies and address practical application challenges. This ensures that the study is grounded in empirical data collected directly from the project, providing a solid foundation for practical application and meaningful insights. The data used include project documentation, design information, TBM-generated data and tunnel segment production data. The iterative DSR process facilitated the design, implementation and evaluation of a digital twin prototype, ensuring alignment with the project's specific requirements. The research design followed the key phases of the DSR methodology (Figure 1):

(1) *Problem identification and motivation:* The initial phase involved identifying the challenges associated with integrating BIM and digital twin technologies in tunnelling projects. This phase included a detailed analysis of existing processes and data structures to establish the need for a tailored digital twin solution.

(2) *Objective definition and data collection:* This phase aimed to define the specific objectives of the digital twin framework. Data collection focused on acquiring comprehensive datasets essential for understanding the tunnelling processes and technological requirements. The data encompassed construction workflows, design models and sensor data from TBMs, forming the foundation for framework development.

(3) *Design and development:* The third phase focused on designing the digital twin framework. This included structuring data for integration, ensuring consistency and developing algorithms to facilitate seamless data integration. Prototypes of the digital twin were iteratively refined based on feedback and operational testing.

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Figure 1. Research design

(4) *Demonstration:* The digital twin was implemented within the project environment to evaluate its functionality and utility. This phase involved testing the digital twin against defined use cases, such as supporting design validation and progress performance monitoring using the generated as-built model.

(5) *Iteration and refinement:* Based on insights from the demonstration phase, iterative improvements were made to enhance the digital twin framework. This included addressing any identified limitations, optimizing algorithms and improving data integration processes to better align with project requirements.

(6) *Evaluation:* The evaluation phase involved analyzing the performance and effectiveness of the digital twin framework. Metrics included efficiency gains in the BIM-digital twin modelling time, usability and impact on decision-making processes. The results informed adjustments to optimize the digital twin's performance and relevance.

The final analysis and conclusions phase helps synthesize the findings from all previous steps to evaluate what worked well and what did not. Lessons learned were documented to inform future research and practice, and results were contrasted with existing literature to highlight contributions to the state of the art, as well as implications for practice and future research.

The subsequent sections of the paper are structured to reflect this framework, detailing the processes, challenges and outcomes encountered throughout the DSR cycle. The findings offer insights into the socio-technical dynamics of BIM and digital twin implementation and contribute to advancing knowledge on their integration in complex construction projects.

3. Related literature: BIM and digital twin in tunnelling construction

BIM research for underground construction has traditionally been less developed than that for above-ground structures (Chapman *et al*., 2020). Nevertheless, in recent years, there has been growing interest in researching how BIM and digital twin technologies can support complex infrastructure projects. Research has progressed from utilizing the Industry Foundation Classes (IFC) schema to model individual components to adopting more holistic approaches. These advanced methodologies now encompass not only the tunnel structure itself but also the

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surrounding elements within the underground environment, providing a more comprehensive representation of the tunnel system (Yabuki, 2009; Yabuki *et al*., 2013).

Borrmann and Jubierre (2013) proposed a detailed shield tunnel product model complemented by a coherent multi-scale representation of semantics and geometry for visualization in IFC-based viewers. Aiming to extend the IFC schema, Vilgertshofer *et al*. (2016) introduced a concept to enable shield tunnel product representation based on three consecutive models of extension. Expanding beyond tunnelling products and components, Zhou *et al*. (2018) proposed a framework to include segment assembly operations into the IFC schema. The authors proposed using an algorithm to automatically identify design information required to allow linking different segment attributes in the model, proposing the application of IFC within parametric design software like Dynamo and Grasshopper for fast modelling and design optimization. In the same direction, Hegemann *et al*. (2012) explored extending and linking the IFC class hierarchy to mechanized tunnelling, particularly by considering the compatibility of IFC classes with TBMs. Similarly, Zhong *et al*. (2018) presented a BIM-based shield tunnel structure information model using an extended IFC description.

An important aspect of tunnelling is considering the relationships with the surroundings and underground conditions. In this area, and taking a holistic view, Koch *et al*. (2017) explored the integration of data from four different data models in mechanized tunnelling construction: a ground model, a boring machine model, a tunnel lining model and a built environment model. The four different models were linked through an open IFC environment. The resulting model was tested to verify its use in selected case studies. Some limitations of the proposed approach concerned visualization, which could be cumbersome when implemented on larger projects requiring detailed geometry. Furthermore, the model limited the scope of information that can be exchanged with the IFC protocol. Similarly, in related research, Ninic *et al*. (2020, 2019, 2017) introduced SATBIM, a concept for parametric modelling to assess the safety and stability aspects of tunnelling design. The concept enables the parametric representation of key system components such as soil excavation, tunnel lining and TBMs, within an information model structured for different levels of details (LoDs).

Specifically for mechanized tunnel construction, Hegemann *et al*. (2020) presented a BIM data integration model to derive as-built documentation of segmental lining rings. In addition to aligning the BIM model to the as-is position, the proposed model integrates data such as thrust cylinder loading history, operational and grouting pressures and segment lifecycle documentation. Similarly, Zhu *et al*. (2020) proposed a concept for the automatic generation of a tunnel information model leveraging BIM functionalities and visual programming language to manipulate BIM elements and process data. The authors validated the model by successfully transferring multiple design alternatives of a tunnel project into a comprehensive tunnel information model. Alsahly *et al*. (2020) explored the integration of BIM and finite element (FE) simulation in mechanised tunnelling, aiming to automatically extract important information such as geology, alignment, lining and process parameters, which are required to perform FE simulations and analysis of the tunnel drive. Getuli *et al*. (2021) presented a conceptual framework for the generation of as-built drawings combining BIM models and the information created from the TBM. The authors aimed to address the challenges posed by complex curved geometries by modelling curves' segment parameters and automating the process through four interlinked algorithms. While the authors demonstrated the feasibility of their approach, the framework was not tested or validated in a real project setting, leaving its practical applicability unproven.

In another stream of research, some authors have explored the use of scanning devices for automated as-built model generation. Cheng *et al*. (2019) proposed a general approach to automatically identify the different types of components present in tunnels (e.g. rails, cross-sections, pipes, catenary equipment and refuges) and create parametric as-is BIM supported by terrestrial laser scanning data. Arastounia (2016) proposed an automated method for as-built model generation in the context of subway tunnels using mobile Light Detection and Ranging data. Similarly, Xue *et al*. (2021) proposed a concept for a metro tunnel 3D reconstruction based on structure from motion (SfM) and direct linear transformation.

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In the transition from BIM to digital twins, researchers have conceptualized the technology from diverse perspectives and for a range of applications. Yu *et al*. (2021) proposed a decision analysis framework for tunnel operation and maintenance based on digital twin technology. The framework introduces an organizational approach for tunnel twin data, which builds upon the extended COBie standard, and employs semantic web technologies to integrate data, objects and knowledge. This approach was tested and validated using a case study of a road tunnel in China. The authors further developed their concept to facilitate knowledge-driven decision support for tunnel electromechanical equipment maintenance. Likewise, Wu *et al.* (2022) explored the use of digital twins to monitor tunnel operations and traffic. The authors focused on constructing a tunnel digital twin and implementing virtual-real fusion operations through 3D video fusion. Tijs (2020) analyzed the prerequisites and structure of a digital twin for tunnels in the Netherlands. The author emphasized the significance of incorporating a digital twin into tunnel operations and maintenance based on demand analysis.

Zhao *et al*. (2022) introduced a digital twin model tailored for intelligent safety assessment in tunnel construction. The authors used real-space information collection technologies, including the Internet of Things (IoT) and intelligent sensor technology, to gather data from ongoing projects. The virtual twin model was created by integrating BIM technology and machine learning algorithms. The effectiveness of this approach was demonstrated through an example involving the settlement of the arch crown during tunnel construction. Sepasgozar (2020) explored virtual and augmented reality, as well as digital twin applications. The author developed and implemented an augmented reality system for construction, including Piling AR (PAR) and a virtual tunnel boring machine (VTBM) module. VTBM, designed as a multiplayer module with avatars, allows users to experience the operation of a tunnel boring machine. The study proposed using immersive construction modules and digital twins for educational purposes. Shen *et al*. (2022) explored the use of a digital-twin-based framework to improve the tunnel lighting environment. The study included the design of a virtual world through VR. The authors also explored numerical simulations and field experiments to enhance the design of the luminaires of the tunnel.

Salzgeber *et al*. (2024) examine the transition from digital models to digital twin in tunnel construction, emphasizing the critical role of structured, real-time data synchronization. They highlight the need to address both technical and non-technical barriers, such as data quality, ownership and collaborative management practices, to enable effective digital twin implementation. Although the authors propose a framework for tunnel twin development, it remains unvalidated in a real-world context, leaving a gap in practical application evidence.

In summary, research on the integration of BIM and digital twin technology in infrastructure projects has been steadily growing. There is a consensus that the full potential of BIM and digital twin applications in tunnelling construction has not yet been fully realized, and current research efforts are still in their early stages (Costin *et al*., 2018). Despite notable advancements in recent years, mechanized tunnelling digital modelling research has predominantly centered around conceptual frameworks rather than practical applications. The authors acknowledge that it is not easy to find facilities that possess comprehensive as-built relevant data beyond a 3D representation of distinct building components. There is a primary need to establish a standardized method for BIM modelling in tunnel projects during both the design and construction phases (Getuli *et al*., 2021).

The gaps in the transition from BIM to digital twin are similar. Current research is mainly centered on broad technologies, including conceptual frameworks and general modelling methods, with a particular focus on the operation and maintenance phase (Parn *et al*., 2024). Specifically, digital twin applications to support the design and construction phases in mechanized construction using TBMs remain underdeveloped. While studies such as Zhao *et al*. (2022) and Wu *et al*. (2022) explore innovative techniques for digital twin development, including real-space information collection and video fusion, they do not include in their methodologies the vast amount of data generated by TBMs. This is a critical gap, as TBM data is essential for optimizing tunnelling processes, enhancing decision-making and ensuring

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early integration of digital twins in mechanized tunnelling construction. In fact, a recent comprehensive literature review of the use of digital twins for intelligent tunnel construction highlighted that digital twin "still requires further development from concept to practice" (Li *et al*., 2024).

Furthermore, the lack of documentation of meaningful digital twins implemented in projects also affects our understanding of how digital twins can be effectively integrated into complex socio-technical systems, such as those involved in mechanized tunnelling construction. Beyond the technological challenges of developing a digital twin as a functional product, it is equally critical to examine how it can be seamlessly integrated into existing processes and operations to support project planning and control using lean construction methods and tools such as the Last Planner System® (Ballard, 2000). This study aims to address these gaps by analyzing not only the technical challenges involved in developing a digital twin but also its practical application in enhancing design and construction workflows.

4. Problem identification and objectives

The research involves the implementation of a digital twin system within the context of a 2.4 km tunnel construction project in France using a variable density TBM. Recognizing the wealth of available data, the project team set the objective of integrating these resources to develop a digital twin system to support operations. Analyzing the construction process and the data from various systems provided the foundation for defining the requirements of the intended digital twin. The technical approach to design the digital twin (e.g. data model and algorithms) aimed to address three specific operational challenges and use cases:

(1) *Automated generation of the tunnelling lining segment status model (as-built):* The TBM should follow a trajectory designed based on geo-exploration data to avoid unstable strata and protect underground resources. However, deviations from the target path can occur due to unbalanced loads in mixed geological conditions (Liu *et al*., 2019). Therefore, the final location of the segments being automatically installed by the TBM system may differ from the as-plan model. Accordingly, the final position of the segments needs to be adjusted to derive an accurate as-built model. In modern TBMs (as in the case of this project), data about the final position of the segments can be extracted from TBM steering software. Nevertheless, this position may not be accurate in some cases due to deviations in the planned trajectory axis and general measurement inaccuracies. Therefore, the final segment position must be confirmed via additional surveying activities on the field. The data from both, TBM steering software and surveying confirmation should be combined to derive an accurate as-built position of the segments. The project team aimed to automate this process by integrating positioning data from both sources into the BIM model.

(2) *Traceability of segment manufacturing information:* beyond generating the 3D representation of the as-built tunnel, the project team aimed to enrich the model with relevant pre-cast segment manufacturing data as a starting point for the digital twin creation. Tunnel segments are pre-casted outside the project in a controlled manufacturing environment. The quality control in this production process allows the individualization of each segment via serial numbers. The segments are transported to the construction site to be finally installed using the automated TBM systems. Attached to each segment serial number there is important segment production data such as segment type, production date, concrete batch, reinforcement design, etc. Incorporating this data into the BIM model generates an additional layer of valuable information which can be leveraged in future phases of the tunnel lifecycle. For instance, when there is a quality issue/request related to an individual segment, the information, including manufacturing data, could be retrieved through the model using a unique ID (Figure 2 left).

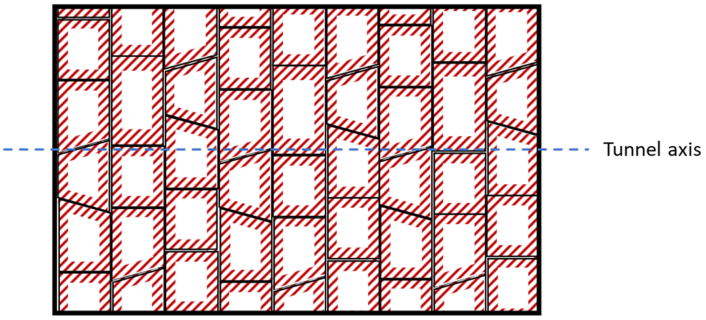
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Figure 2. Typical segment production ID used in production (left), segment storage (right)

(3) *Facilitate tunnel bracketry design and production workflows:* an additional practical application of having the segments' manufacturing information available in the model is related to planning and installation of bracketry to support mechanical, electrical and overhead line equipment. After the lining of segments is done, the completion of the tunnel requires intense drilling on the segments to anchor the bracketry for supporting facilities. One of the main challenges of this phase is related to de definition of "no-drill zones." The structural integrity of the tunnel relies on the segments. Therefore, defining the zones where drilling can compromise reinforcement is key to ensuring tunnel stability and functionality. It is common practice to indicate no-drill zones in the model. However, the segments as-built position is only known after installation. Additionally, the tunnel lining consists of rings containing various types of segments (e.g. key, counter key, standard), creating a complex array that complicates the definition of an efficient linear drilling configuration (Figure 3). Therefore, an as-built model including these no-drill zones can facilitate both design and construction activities. In practical terms, this model can then be utilized by mechanical, electrical and overhead line equipment designers through automated clash detection to establish a safe and efficient bracketry system, further contributing to the development of the digital twin. Furthermore, installation and drilling teams can also use the model to navigate around no-drill zones efficiently during construction. Considering these benefits, the project team aimed to develop an as-built model that clearly defines these zones. Figure 3, adapted from Douglas *et al*. (2019), provides an illustrative view of the tunnel lining on a plane, highlighting in red pattern fill the substantial and complex surface area taken up by no-drill zones.



Source(s): Based on Douglas *et al.* (2019)

Figure 3. Representative view of the tunnel lining highlighting "no-drill zones"

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5. Digital twin design and development

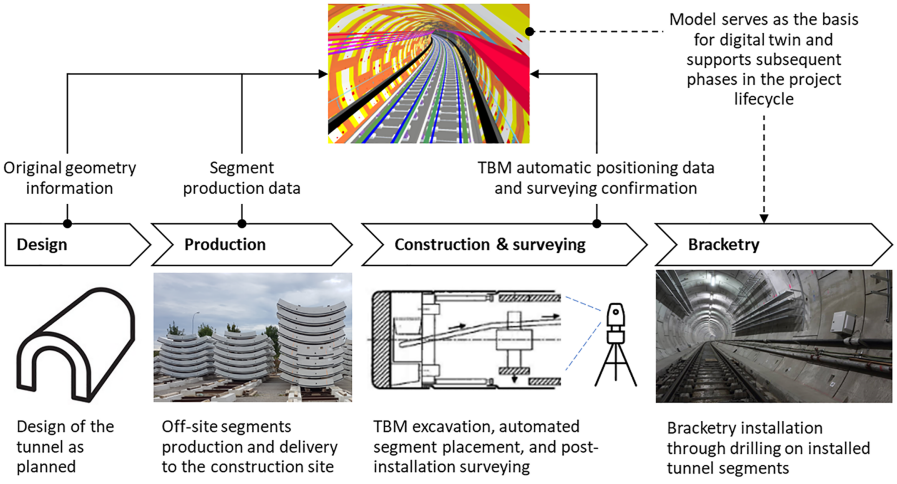
Figure 4 schematically illustrates the data integration concept developed to generate the digital twin. Original tunnel geometry and as-planned information are derived from the tunnel lining design and stored as Project Intent Information (PII). Segment production details, along with identification numbers, are sourced from the segment manufacturer's proprietary software and are stored with the Project Status Information (PSI). Final segment positioning is established by integrating TBM positioning data with survey-based verification. Ultimately, the generated digital twin is intended to support future activities, particularly in bracketry design and construction.

5.1 Data sources

Figure 5 provides an overview of the different proprietary systems and their integration to derive the digital twin. To different degrees, data sources had to be structured and organized in a machine-readable format to be integrated using algorithms developed in Dynamo. Details of the data sources and the work done to prepare the data are as follows:

TBM data: All data from the segment production and the tunnel excavation using the TBM is collected in TPC (Tunnel Process Control). TPC is a specialized software solution designed to monitor and manage tunnelling operations in real time. The software integrates data from TBMs, surveying instruments and other construction equipment, allowing for continuous oversight of excavation progress, segment placement and alignment (Tunnelsoft, 2024). In this study, all data from the segment production (e.g. production date, segment ID, type) was collected within TPC. Since the TBM scans the tunnel segment before installing it, there are records about the time and place where each tunnel segment was installed.

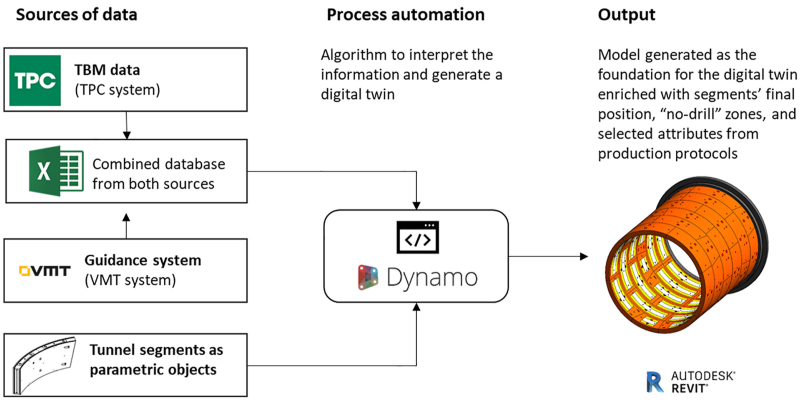
Guiding system (VMT): TBMs commonly employ advanced guidance systems to achieve millimeter-level precision, even where tunnels extend over several kilometers. These systems continuously calculate real-time data essential for maintaining alignment with the planned tunnel axis. Data from the guidance system was used to validate and, when necessary, correct information obtained from the TBM's TPC system. This validation was crucial for identifying measurement deviations caused by factors such as rolling (lateral shifts of the TBM against the tunnel face) and misalignments from the planned trajectory.



Source(s): Authors' own work, combined with TBM and tunnel lining schematics adapted from Maidl *et al.* (2012)

Figure 4. Data integration concept

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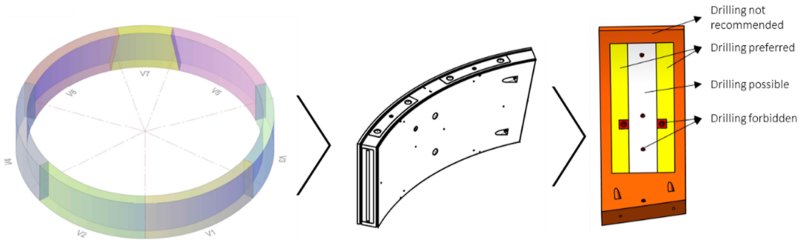
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Figure 5. Data integration through Dynamo script

Combined Excel database: For the scope of this project, an MS Excel table was used to consolidate data from the TBM and the guidance system. The ring position (centre point) in the MS Excel originates from the TBM's TPC system and was corrected using information from the VMT guidance system. Additionally, manual measurements were performed to increase accuracy and confirm the centre point positions. This MS Excel database was specifically designed for the needs of this project and can effectively function as a basic database for its scale. Nevertheless, for more complex applications the use of a genuine relational database management system is recommended, given their structural organization, relational integrity and robust data manipulation capabilities.

Tunnel segments split into drill zones: Each ring in the project consists of seven tunnel segments (see Figure 6, left, V1 to V7). From the design phase, the detailed geometry of each tunnel segment was already available in a 3D format. To be able to automatically generate the as-built model, smart parametric objects were required. Parametric objects allow to change their geometric properties based on input parameters. The geometry of the object is defined by parameters and logical relationships between these parameters. This means that changes to one parameter will automatically adjust other related features in the object, maintaining consistent relationships.

The basic geometry from the design phase had to be further detailed to identify the desired drilling zones that will support the automated drilling clash detection and installation of bracketry (Figure 6, right). This process also included taking a deeper look into each segment's



Source(s): Authors' own work

Figure 6. Preparation of segment object families

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structural details along with tunnel lining design to verify the location critical reinforcement that must be avoided in the bracketry drilling process. These zones were defined considering segment structural integrity and were classified by the project team as drilling not recommended, drilling preferred, drilling possible and drilling forbidden.

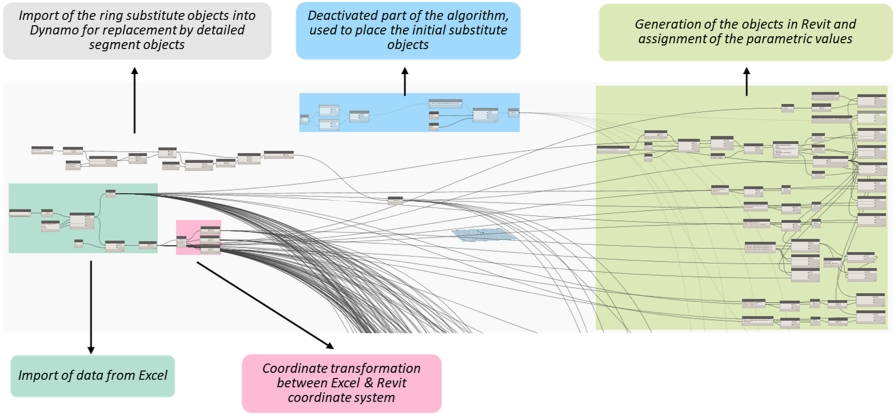
The parametric objects for the tunnel segments were created in Autodesk Revit using Dynamo. Segment parameters and their relationships were defined to allow segment rotation around a predefined tunnel axis (the as-is axis as captured from TPC and VTM systems). This is necessary to specify the keystone position correctly after placing the object. The parametric objects are also prepared to hold the information specified by the client (e.g. classification structure, segment ID).

6. Demonstration and refinement

An innovative algorithm was developed in Dynamo to generate the PSI (as-built model) based on the consolidated data from the various monitoring sources. Dynamo is an open-source visual programming platform that is widely used in the field of BIM, particularly with Revit applications (Dynamo BIM, 2021). Dynamo provides a graphical interface for creating custom scripts and workflows, enabling task automation and efficient control and modification of BIM elements in Revit.

In Figure 7, the developed algorithm performs a set of logical steps to create the digital twin: it reads data from the MS Excel database, places tunnel segments according to as-is coordinates, adjusts the object parameters according to the updated geometry and appends additional information (e.g. ring ID and manufacturing data).

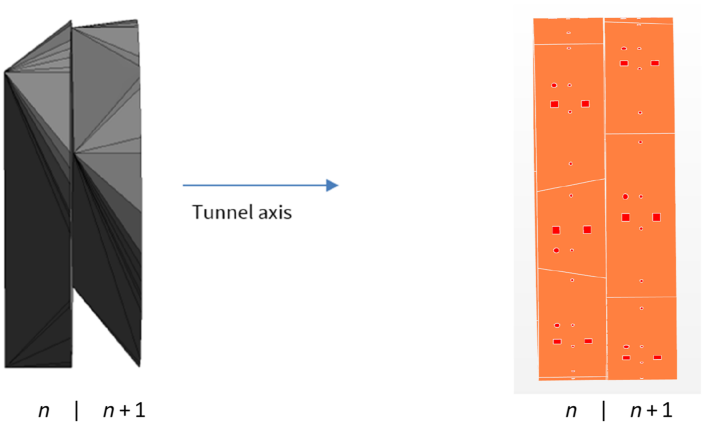
As emphasized in previous research, the technical challenges associated with developing a digital twin as a functional product often derive from the complexity and volume of data to be processed, which can overwhelm data processing systems (Getuli *et al*., 2021). To address these challenges and streamline the data processing required for digital twin generation, the algorithm is designed to incrementally load information into the model, beginning with the basic geometry. Specifically, the algorithm utilizes a ring substitute object (Figure 8, left), which represents the front and back planes of each ring. By leveraging this approach, the algorithm dynamically adjusts the position of each substitute ring based on the corrected real centre point coordinates, ensuring both accurate and efficient model generation. The ring substitute serves as an optimization mechanism to mitigate excessive computational demands within the algorithm. It is initially positioned in the model using a minimal data set to



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Figure 7. Overview of the Dynamo algorithm used

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Figure 8. Two-ring substitute objects with adjoining planes (backplane n and front plane n+1)

streamline processing efficiency. Once the positional accuracy is validated and any necessary corrections are applied, the algorithm dynamically integrates the full dataset into the model, ensuring precise spatial alignment and continuity within the digital twin framework.

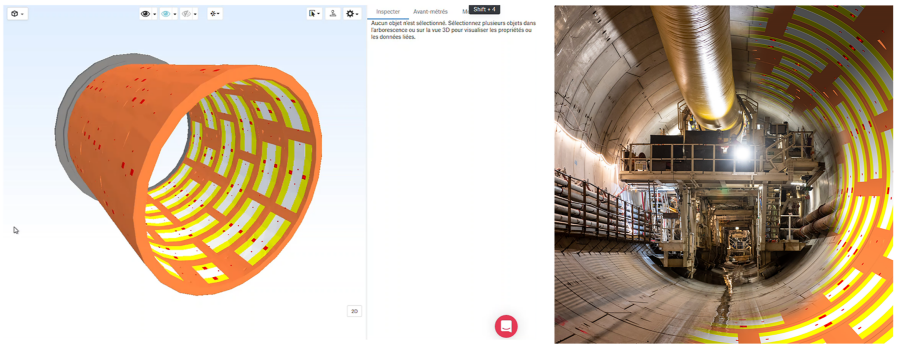
In the next step, the algorithm models the individual rings (V1 to V7) within the ring assembly, beginning with the Keystone. During this process, non-parallel adjoining planes were identified, as shown in Figure 8 (right). Specifically, the backplane of the nth element is not parallel to the front plane of the (n+1)th element, resulting in slight separations between adjacent segments at the top and minor overlaps at the bottom. To correct these deviations, the algorithm utilizes validated positioning data from the site. This adjustment process is fully automated in Dynamo, where data gathered during the surveying process is linked to the unique ID of each segment. This ensures that all adjoining ring planes are aligned parallel to one another, based on the definitive position of each ring. The accuracy of the model is verified through cross-checks with onsite surveying measurements, ensuring precision and reliability.

The final step is replacing each substitute element with the corresponding parametric tunnel segment object, labelled V1 to V7, as shown in Figure 6. Segment types and additional production information—such as production and casting dates and ring numbers—are sourced from the MS Excel database (refer to Figure 5) and incorporated into each tunnel segment. This approach not only enhances accuracy but also significantly reduces the time and effort needed in model generation, ensuring that the as-built BIM model is a precise representation of the actual construction.

6.1 Results and evaluation

Figure 9 shows a 3D visualization of the model in an IFC viewer (left) and a superposition of the tunnel segments with a picture from the tunnel under construction (right). The final digital twin of the tunnel segment lining provided an accurate representation of what had been built at each stage. The model successfully integrates and displays data derived from previous phases in the project life cycle, such as the original BIM model and key manufacturing data from tunnel segment production. The model also integrates the "no-drill" zones, supporting the design and execution of future project phases. In this case, the model is used to optimize the design and construction of the bracketry system for mechanical, electrical and overhead line equipment and to perform clash detection with segment reinforcement. The availability of the model with this level of accuracy and information can also support additional use cases, both during construction and later for operations and maintenance.

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Figure 9. Final visualization of the BIM/digital twin model in IFC viewer

To verify the model's accuracy, we compared its results with manual surveying measurements taken for each ring. This review identified minor deviations, which we manually adjusted in the model. However, these discrepancies did not result from errors in the algorithm but from differences between the TBM location data and manual surveying measurements. Such variations are normal in tunnelling construction since installed rings can shift slightly due to natural geological accommodation.

The algorithm significantly improved the efficiency of data consolidation and digital twin creation. Manually consolidating information and updating the model would have taken 15–20 minutes per ring, while the algorithm reduced this to just 1 minute of manual quality checks. This improvement cut cycle time by more than 90% per ring, making the process far more efficient and less labor-intensive. Additionally, with the required adjustments, the algorithm can be implemented in future projects.

7. Discussion

This case study makes an important contribution to the fields of BIM and digital twin systems for construction by applying computational design methods to integrate TBM and segment manufacturing data into an as-built BIM model, serving as the foundational framework for a forthcoming digital twin. In line with the objectives of the study, the authors analyzed the findings and experience of the study from a broader perspective in relation to literature considering the digital twin as a functional product and its use to facilitate design and lean construction workflows.

7.1 From BIM to a functional digital twin

Boje *et al.* (2020) highlight that within the interconnected relationship between the physical and digital components of a digital twin, BIM is often regarded as a foundational element. This is because the development of a digital twin typically originates from BIM, which initially serves as the digital representation of the asset. As the project progresses, BIM is incrementally enriched with real-time data streams, transitioning into a fully operational digital twin capable of supporting real-time monitoring and decision-making. However, the vast volume of real-time data collected from multiple sensor sources in TBM construction presents a significant challenge. Traditional data processing systems often struggle to manage and interpret this influx of information, limiting the ability to derive actionable insights (Getuli *et al.*, 2021). A promising approach to address these data management challenges is the integration of parametric design tools, such as Dynamo and Grasshopper, into the DT framework. Zhou *et al.* (2018) suggest that these tools can facilitate automated tunnel model generation with precise segment placement,

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thereby improving design accuracy and construction efficiency. The innovative computational design method applied in this study builds upon this idea and the successful integration achieved validate the potential of parametric design for advancing BIM and digital twin applications in complex projects, improving both efficiency and precision in digital modelling workflows. Nevertheless, certain manual steps were necessary to ensure system functionality. This limitation primarily derives from the lack of integration between the various systems involved in tunnelling projects. Multiple stakeholders relying on separate, discrete systems, create more barriers to data sharing and integration. For example, data from tunnel segment production is often restricted to proprietary systems managed by manufacturers. Likewise, TBM data is stored in exclusive systems developed by equipment suppliers. The experience in this case study confirms the view of researchers and industry practitioners that challenges in fully leveraging BIM and digital twin are not just about the technology but span into broader organizational and structural industry issues (Getuli *et al*., 2021; Stascheit *et al*., 2018).

This study also highlights the need for a standardized approach to managing tunnel geometry and information. Tunnels feature complex and often irregular geometries, making it challenging to represent them accurately in IFC—particularly for nonstandard tunnel designs (Gong *et al*., 2024; Luo *et al*., 2022). Starting with Yabuki (2009), several efforts have aimed at establishing a certain degree of standardization (e.g. BuildingSmart, 2020; Lee and Kim, 2011; Stascheit *et al*., 2013). However, essential design information remains unrepresented, or the standard itself has struggled to gain sufficient industry traction for widespread implementation. In this case study, the original tunnel design had to be detailed and parametrized to meet the interoperability requirements necessary to facilitate integration with other data sources. This effort could have been reduced if those BIM-IFC data standards and requirements were set up from the very beginning of the project.

7.2 Digital twin construction – informing and controlling construction operations

Conceptually, construction with digital twin information systems consists of two main components: (a) intent information, encompassing as-designed and as-planned data, such as original BIM models and construction plans, and (b) status information, comprising as-built and as-performed data, including details like final segment positioning (Sacks *et al*., 2020). In this case study, these components are processed separately and eventually brought together to form the digital twin. This generated virtual system proactively informs the physical world, enabling the construction project team to compare the intended design with the actual construction in real time. This approach offers a more meaningful and productive outcome than traditional as-built models, which are typically compiled reactively at the end of the construction phase.

In terms of operational efficiency, the automation gained with the proposed solution contributes to reduce manual work that would be needed to produce the digital twin by manual means otherwise. Per tunnel ring, the cycle time to model a ring was optimized by more than 90%. The benefits extend to further operations in the project lifecycle since the digital twin system has information that can be fed back to inform the ongoing fabrication level design detailing, part of the design intent. Once the segments are placed, the information in the status model is used as inputs for proactive engineering bracketry drilling design. This cyclical flow—integrating information from both intent and status back into the design—represents the core operational value provided by the developed digital twin in this study. Additionally, similar operational benefits can be extrapolated to upstream workflows, such as streamlining the bracket construction process and ensuring that relevant segment production information, including sensors, is linked to its precise location within the built tunnel lining for future maintenance and operations.

7.3 Limitations and future work

The single case study design may limit the generalizability of the findings. The parametric segment objects and data models developed are specifically tailored to the unique project

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context and the proprietary systems utilized. Future research could extend this work by evaluating the adaptability of the proposed framework across different tunnelling projects, exploring its scalability in diverse geotechnical and operational conditions and integrating standardized IFC to enhance interoperability. This research demonstrates efficiency gains in the modelling process and highlights the prompt availability of the digital twin to support lean operations, decision-making and related processes. However, quantifying overall operational savings for the project remains challenging, as there is no benchmark data from a comparable control project executed without a digital twin for direct comparison. Despite these limitations, the authors believe that the overall innovative approach to digital twin generation, along with the practical use cases demonstrating meaningful applications in an operational context, offer valuable insights for researchers and practitioners working in similar settings.

8. Conclusions

This study demonstrates the feasibility and benefits of implementing and using digital twin construction systems and methods to streamline the construction of tunnels with TBMs, as illustrated through a use case in a mechanized tunnelling construction project. The implemented algorithm achieved a 90% reduction in digital twin modelling time while enhancing lean design and construction workflows. By generating a dynamic representation of the tunnel's physical state in short cycles, it provided the project team with timely, data-driven insights to support decision-making and process optimization.

A key insight derived from this experience is that defining and implementing systems with adequate data exchange standards to enable data interoperability among the various software products is essential for realizing the tangible benefits of BIM design and digital twin. Another is that despite imperfect data interoperability across the different systems involved, it is feasible to compile an accurate digital twin status model automatically using state of the art technology and innovative parametric design algorithms. The study also provides insights on how a purpose-built digital twin can support complex construction processes, such as tunnel segment lining and bracketry construction, thus improving the flow of tunnelling operations and avoiding the waste of rework by enabling error-free installation of systems. This is achieved by providing project construction and design teams with accurate information on current conditions in real-time.

Notably, the study establishes a crucial link between research and practical applications, building upon knowledge to formulate a novel approach grounded in a real-world project. Ultimately, this work contributes valuable insights for researchers and industry practitioners seeking to advance digital twin applications in tunnelling and offers a replicable framework that may be adapted for broader construction contexts. From a theoretical standpoint, it enhances our understanding of the positive interactions of BIM (and now DTC) and lean construction principles.

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Corresponding author
Eder Martinez can be contacted at: eder.martinez@implenia.com

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